

AI-tool adoption and self-reported job satisfaction among professional software developers: a descriptive analysis of the Stack Overflow Annual Developer Survey

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Abstract

We use the 2025 Stack Overflow Annual Developer Survey to describe the relationship between professional developers' reported adoption of AI coding tools and their reported job satisfaction, controlling for years of professional experience and primary developer role. In a complete-case sample of 24,354 respondents, reported AI-tool adoption is associated with a 0.151-point higher job-satisfaction score on a 0–10 scale (standard error 0.033, $p < 0.001$). The association is statistically precise but small. The analysis is descriptive: the survey is observational, selection into the survey is not random, and both AI-tool adoption and satisfaction are self-reported. We discuss the limits this design imposes on causal interpretation. This paper is a teaching artefact produced for a seminar on AI for research; it is not intended for publication.

1 Introduction

AI-enabled tools have quickly become part of many software developers' work. Their adoption may coincide with changes in the experience of work: automation could remove repetitive tasks, while unreliable output or pressure to adopt new tools could create frustration. This paper asks a deliberately narrow question: among professional developers responding to the 2025 Stack Overflow Annual Developer Survey, is reported AI-tool adoption associated with reported job satisfaction?

The available data cannot establish whether AI tools change satisfaction. Adoption is not randomly assigned, and the survey observes respondents at only one point in time. We therefore estimate a descriptive conditional association, not a causal effect. The contribution is a transparent worked example for teaching an end-to-end research workflow: define variables, construct an analysis sample, run a reproducible regression, inspect figures, and report the limitations alongside the estimate.

2 Data

The data are from the 2025 Stack Overflow Annual Developer Survey [Stack Overflow, 2025]. The downloaded file contains 49,191 responses and 172 variables. Stack Overflow recruits respondents openly rather than by probability sampling, so the respondents should not be interpreted as a representative sample of all software developers. The data are released under the Open Database License 1.0 and Database Contents License 1.0.

We retain the 37,467 respondents whose main branch identifies them as professional developers. Job satisfaction is the survey's numeric `JobSat` response, measured from 0 to 10. We classify

respondents as AI tool adopters when `AISelect` reports daily, weekly, or monthly/infrequent use; the two “No” responses form the non-adopter group. Professional experience is numeric `WorkExp`. Because `DevType` can list multiple roles, we use its first listed role as the primary role. Complete-case filtering and the exclusion of missing or uncodable AI responses leave 24,354 observations.

3 Method

We estimate ordinary least squares regressions of job satisfaction on an indicator for reported AI-tool adoption, years of professional work experience, and fixed effects for primary developer role. The model includes 31 role indicators; “Developer, full-stack,” the most common role in the analysis sample, is the omitted category. We report conventional OLS standard errors.

The coefficient on AI-tool adoption compares adopters and non-adopters with the same recorded years of experience and primary role. This adjustment addresses only those measured differences. It does not make adoption exogenous and should not be read as the effect of using AI tools.

4 Results

Table 1 reports the central descriptive and regression results. Among the 19,680 reported adopters, mean job satisfaction is 7.236; among the 4,674 non-adopters, it is 7.132. Both groups have a median of 8. Figure 1 shows that the raw difference in means is modest.

After adjustment for experience and primary-role fixed effects, the adoption coefficient is 0.151 satisfaction points (standard error 0.033, $t = 4.63$, $p < 0.001$). The model R^2 is 0.021. Thus, reported adoption has a positive cross-sectional association with satisfaction, but the estimated difference is only about 1.5 percent of the full 0–10 response scale and the model explains little of the individual variation in satisfaction. Respondents in the analysis sample report a mean of 13.6 years and a median of 11 years of professional experience; Figure 2 shows its right-skewed distribution.

Table 1: Job satisfaction and reported AI-tool adoption

Group or estimate	N	Mean	Median
AI tools not used	4,674	7.132	8.00
AI tools used	19,680	7.236	8.00
OLS adoption coefficient: 0.151 (SE 0.033; $p < 0.001$)			
Model $N = 24,354$; $R^2 = 0.021$; experience control and role fixed effects included.			

5 Limitations

The headline limitation is that the estimate is not causal. Selection into the Stack Overflow survey is not random: developers who encounter the survey and choose to complete it may differ systematically from other developers. The sample therefore cannot support population-wide prevalence claims without strong additional assumptions.

Both central variables are self-reported. Respondents may interpret AI-tool use differently or misreport its frequency, and a single satisfaction score is an imperfect measure of work experience. The survey contains no instrument or other source of plausibly exogenous variation in adoption. Its cross-sectional design also prevents us from establishing whether adoption preceded satisfaction.

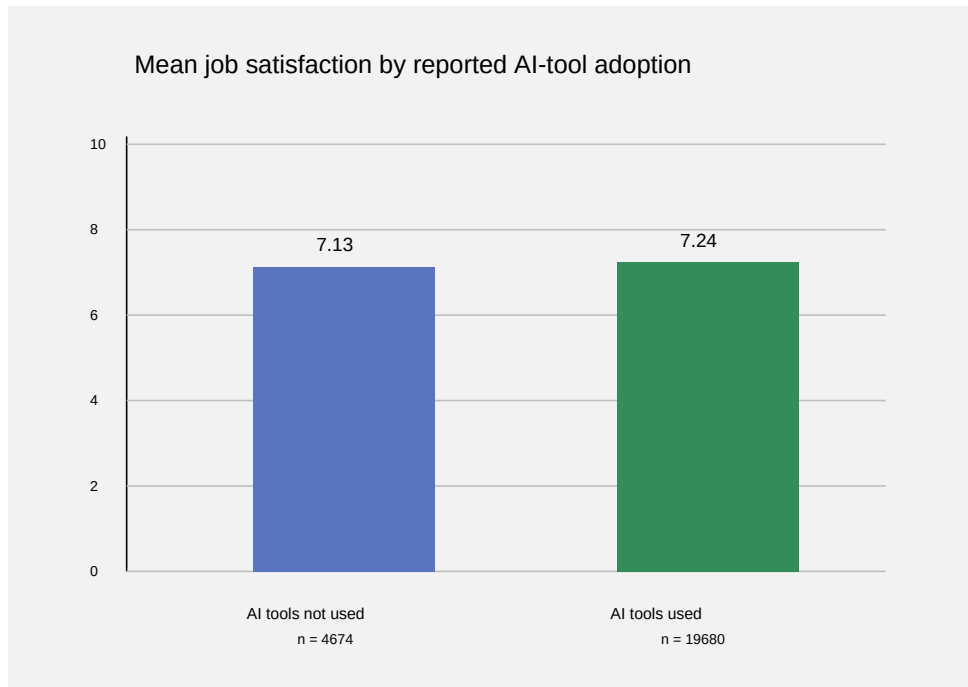


Figure 1: Self-reported job satisfaction by reported AI-tool use, analysis sample.

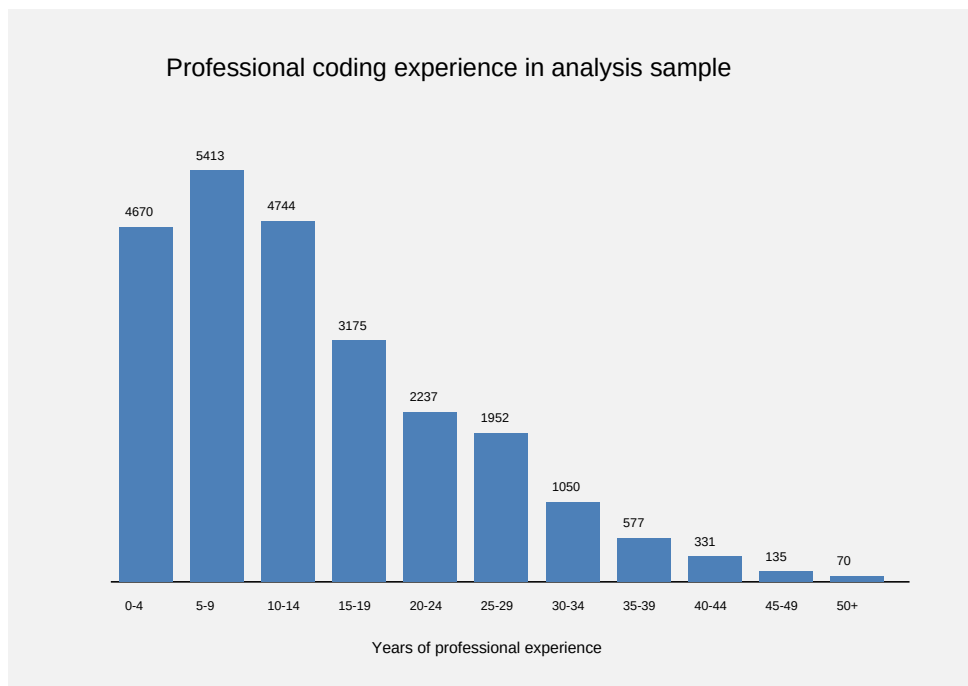


Figure 2: Distribution of years of professional coding experience in the analysis sample.

Finally, the regression controls only for experience and primary role. Omitted factors such as team culture, employer policy, job quality, developer ability, pay, and enthusiasm for new technology may influence both adoption and satisfaction. Any of them could generate or mask the small observed association. The complete-case restriction and our reduction of multi-role responses to the first listed role add further scope for selection and measurement error.

6 Conclusion

Professional developers who report using AI tools also report slightly higher job satisfaction in this survey. The adjusted difference is 0.151 points on a 0–10 scale. We document an association, not evidence that AI-tool adoption causes greater satisfaction; the design cannot distinguish that explanation from selection, reverse ordering, or omitted variables.

This paper is a teaching artefact rather than a publishable empirical study. Its purpose is to demonstrate a reproducible path from a public dataset to an auditable analysis, figures, and a compiled paper while keeping statistical and substantive uncertainty visible.

References

Stack Overflow. Annual developer survey 2025. <https://survey.stackoverflow.co/>, 2025.
Licensed under ODbL 1.0 and DbCL 1.0. Replace year if a different snapshot is used.